

Drivers of Artificial Intelligence Adoption in European Union Enterprises

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Abstract

The emergence of Artificial Intelligence (AI) has greatly transformed the operations of enterprises. Most enterprises across the globe are currently leveraging the explosive powers of AI to enhance their daily operations and optimize performance. This research analyzes the adoption of AI in enterprises across the European Union (EU) 27 in 2024 and the drivers. To accomplish this, we used multivariate statistical methods, particularly factor and cluster analysis, to analyze the trend of adoption across the EU. This led to a discovery of the fact that a gap exists between the EU 27 countries. Digital/energy investments, digital literacy and ICT employment shortages explained the disparity.

1. Introduction

The rapid spread of artificial intelligence (AI) among companies in the European Union has increased the need to better understand the conditions that enable, shape, or limit its adoption. While recent Eurostat data offer detailed insights into how AI is being used across areas such as marketing, production, logistics, and financial management [1], the underlying patterns behind these figures are far from straightforward. As seen in previous phases of digital transformation, the adoption of AI is not driven solely by firm-level decisions. It is also strongly influenced by broader national factors, including labor market structures, levels of digital participation, and investments in technological and energy infrastructure, all of which shape the environment in which firms operate [2] [3] [4]. Considering these factors together is therefore essential to capture the diversity of AI adoption profiles across EU countries.

Because this topic involves many interrelated aspects, looking at one variable at a time does not adequately explain what is really happening behind the observed indicators [5]. A multivariate approach is therefore more suitable, as it allows different AI-related measures and contextual factors move together, helping to identify common patterns and to assess whether countries share similar digital characteristics [6]. This way of analyzing the data follows a long-standing tradition in information systems and digital economy research, where factor analysis and clustering have often been used to understand why and how technology adoption differs across countries [7].

The present study applies this multivariate approach to a dataset constructed from Eurostat indicators to study the adoption of AI technologies in enterprises and its relationship with other indirect indicators. The study uses the 2024 Eurostat indicators datasets on enterprise use of artificial intelligence because timeliness and currency are key when analyzing rapidly evolving technologies. The 2024 data capture important recent shifts in diffusion dynamics and sectoral patterns that earlier years do not reflect [8]. This study has two main objectives: to identify the underlying dimensions that structure the relationships between the variables, to present a concise view of the key drivers and barriers to AI adoption by enterprises and to explore whether EU countries can be grouped into distinct clusters that reflect different digital profiles.

By linking AI use indicators to broader socioeconomic and digital infrastructure factors, this study provides a clearer picture of what drives AI adoption in the European Union. The results offer an empirical basis to characterize cross-country differences and to discuss the digital capacities and challenges that accompany the ongoing diffusion of AI technologies. Therefore, regardless of the number of distinct factors that are produced from the analysis, the scope of our study will only be on the relationship between the factor(s) containing the AI adoption indicators and the other factors not containing any AI indicator (if it exists).

The analysis intends to provide a data-backed answer to the following questions: (1) Are advanced digital economies always high in AI use, or not? (2) How do EU countries cluster base on their level of enterprise AI adoption? (3) Are there geographic or regional patterns in AI adoption among EU Member States? In answering these questions and some others, the research is organized as follows: Sections 1 to 3 introduce the topic, providing further context in Global AI Adoption and Measuring AI Adoption in EU Countries; Section 4 presents the Methodology we

adopted to analyze the AI adoption rate; Section 5 includes the results of the analysis undertaken on the data collected, using factor and cluster analysis, and comparison between all the other factors and the AI factor. Section 6 presents the discussion of the findings, while Section 7 presents the conclusions and implications. All consulted references and results of computational procedures are provided as well.

2. Global AI Adoption

Recently, the global diffusion of AI has accelerated due to the advancement of technologies, increased data accessibility, and institutions are being pushed to adapt to digital changes. In the EU context, indicators point to a consistent growth in the use of AI tools mainly machine learning, natural language processing and robotic process automation in industries like finance, manufacturing, and logistics [1]. This trend is consistent with international evidence showing that companies adopting digital and data-intensive technologies often become more productive and organizationally agile [2].

Outside of the EU, AI has also gained significant adoption in North America and East Asia, driven by strong private sector investment and highly developed digital environment. Countries such as the United States, Canada, China, South Korea, and Singapore demonstrate rapid scaling of AI in cloud services, e-commerce, smart manufacturing, and public administration. Emerging economies in Latin America, the Middle East and Sub-Saharan Africa show increasing adoption as well, often due to efficiency pressures, mobile-first markets, and the growing availability of open-source tools and cloud-based AI services. Although adoption levels vary, the global pattern points to a widening integration of AI in productivity enhancing and decision support processes, consistent with evidence on the organizational gains due to data-driven decision making [3]. Overall, global AI adoption reflects not only technological progress but also a structural transition toward data driven economies.

3. Measuring AI Adoption in EU-27

Although humans have always sought to build thinking machines and self-instructible devices. Alan Turing and several other outstanding scientists made significant strides in the field of artificial intelligence. These previous works greatly aided the modern state of AI. When OpenAI released the first version of Generative Pretrained Transformers (GPT), it quickly gained traction and made that ancient dream looks more feasible and achievable [9].

Since recent times of technological advancement and modernization, the quest for adoption of AI and its accurate usage has always been a front runner in this race. In April 2025, the EU published an AI Continent Action Plan which aims to make the continent a global leader in AI. This plan focuses on developing AI technologies that would enhance the competitiveness of enterprises in EU member states while maintaining democratic values and data security. To measure this adoption, it is pertinent to consider other non-technological indicators that might be affecting the adoption and usage of AI technologies in EU-27.

We focused our analysis on the AI adoptions for marketing or sales, for production processes, for logistics and for accounting, controlling or finance

management [10]. This functional delineation is consistent with international evidence from the OECD, which shows that AI diffusion within firms is task-specific and most prevalent in domains involving customer analytics, production optimization, supply-chain coordination, and financial process automation. Focusing on these four domains therefore provides a methodologically coherent and empirically grounded basis for assessing AI adoption, ensuring alignment with both EU statistical standards and the broader literature on digital transformation in enterprises.

In addition to AI-specific variables, the research also uses a group of non-AI indicators that capture structural, organizational, and macroeconomic conditions known to influence the capacity of enterprises to adopt advanced digital technologies. These indicators are related to ICT specialist recruitment, such as applicants' lack of relevant ICT qualifications, high salary expectations [11]. We also used an indicator that shows whether enterprises employed or attempted to recruit ICT-skilled personnel, this reflect labor-market constraints and organizational readiness that is identified as significant determinants of digital transformation [12]. Furthermore, macro-level expenditure indicators, consumption expenditure on electricity and fuels, information and communication services, and education, serve as proxies for national investment patterns and resource environments that condition enterprise-level technology uptake [13] [10] [14]. Variables such as the total unemployment rate and the share of individuals using the internet to search for information about goods and services capture broader socioeconomic contexts, including digital maturity, labor-force participation, and consumer-driven digital behavior [15].

Incorporating these non-technological indicators is therefore methodologically justified, as they represent organizational and environmental determinants consistent with the Technology–Organization–Environment (TOE) framework and with EU digital monitoring methodologies, both of which emphasize that AI adoption is shaped by interrelated human-capital, infrastructural, and economic factors rather than technological capabilities alone. Nonetheless, this selection of indicators should be interpreted as a pragmatic representation of the complete spectrum of factors shaping AI adoption in enterprises. Rather, our aim is to construct a methodologically coherent and empirically grounded measurement framework that reflects the most relevant and comparable indicators available across the EU-27. Given the constraints of cross-national data availability, the variables included, spanning AI usage in marketing and sales, production processes, logistics, and financial management, alongside the broader organizational, labor-market, and macroeconomic determinants, offer a sufficiently comprehensive representation of the structural conditions influencing AI uptake. While additional micro-level behavioral, sector-specific, or organizational culture indicators could further enrich the analysis, the chosen set provides a robust and policy-aligned basis for assessing variation in AI adoption and digital development across member states.

4. Methodology

This section focuses on providing clarity on the analytical procedures used to investigate the drivers of Artificial Intelligence adoption in EU countries. To uncover patterns and perform grouping among the 27 EU countries, we use two different techniques. First, applied Factor Analysis to reduce dimensionality and extract latent

factors from the dataset. Factor Analysis was chosen over Principal Component Analysis (PCA) because our primary goal was to identify and interpret underlying constructs factors that explain the correlations within observed variables, rather than maximizing variance, which wouldn't provide any insight. Then, we used the factor scores as inputs for hierarchical clustering, enabling us to identify distinct country clusters based on digital and socioeconomic profiles.

4.1. Data

The empirical analysis is based on 2024 Eurostat datasets covering the 27 European Union (EU) states, from those we selected 12 variables, selected for their relevance to enterprise-level AI adoption and their alignment with OECD and EU recommendations. These variables capture both direct measures of AI use (e.g., for marketing, production, logistics, and finance) and contextual factors such as ICT employment, recruitment difficulties, macroeconomic expenditures (energy, ICT, education), unemployment, and digital participation (internet use). All variables are expressed as percentages, either of enterprises or of the population, ensuring comparability across countries.

The dataset includes 27 country-level observations and contains a limited number of missing values (nine missing entries across several variables). As the amount of missing data was small, missing values were addressed using mean imputation. Specifically, each missing value was replaced by the average of the corresponding variable calculated from the available observations. This procedure made it possible to retain all countries in the analysis and avoid unnecessary loss of information. While mean imputation may lead to a slight reduction in variability and may affect correlations, it is widely regarded as an appropriate approach for exploratory multivariate analysis when the proportion of missing data is low [16].

Following the framework, dividing the dataset of the indicators for measuring the adoption of AI, we used 12 succinct variables that are consistent with recommendations from OECD and the European Union. All the values of the variables were obtained from 2024 Eurostat datasets. The fact that all variables were obtained from official entities, and possess a high degree of currency and recency, guarantees that the results of the analysis enjoy a high degree of reliability. [Table 1](#) shows the variables collected. Potential outliers were assessed using box-and-whisker plots and the interquartile range (IQR) criterion. Given the country-level nature of the data, the most extreme values were interpreted as reflecting genuine structural differences between EU-27 rather than data errors. Therefore, outliers were retained in the dataset, as they carry substantive economic and policy-relevant information, particularly regarding digital adoption, labor market conditions, and sectoral specialization [5].

Table 1:

Abbreviations, full variable names and support.

ID	Code	Variable	Support
1	aiMrk	percentage of enterprises that use AI technologies for marketing or sales	(1)
2	aiProd	percentage of enterprises that use AI technologies for production processes	(1)
3	aiLog	percentage of enterprises that use AI technologies for logistics	(1)
4	aiFin	percentage of enterprises that use AI technologies for accounting, controlling or finance management	(1)
5	diffQual	percentage of enterprises that have difficulty to recruit ICT specialists due to Applicants' lack of relevant ICT qualifications from education and/or training	(2)
6	diffSal	percentage of enterprises that have difficulty to recruit ICT specialists due to applicants' salary expectations too high	(2)
7	ictEmp	percentage of enterprises that employed ICT specialists or recruited/tried to recruit personnel for jobs requiring ICT specialist skills	(2)
8	conEnergy	percentage of total population consumption expenditure for electricity, gas and other fuels (% of GDP)	(3)
9	conICT	percentage of total population consumption expenditure for Information and communication (% of GDP)	(3)
10	conEdu	percentage of total population consumption expenditure for Education services (% of GDP)	(4)
11	unemp	Total unemployment rate	(5)
12	intUse	percentage of individuals by total population using the internet for finding information about goods and services	(6)

Support:

(1) Eurostat. Artificial intelligence by size class of enterprise (isoc_eb_ai). Source [here](#).

(2) Eurostat. Integration with customers/suppliers, supply chain management by size class of enterprise (isoc_eb_ics). Source [here](#).

(3) Eurostat. Digital Intensity by size class of enterprise (isoc_e_dii). Source [here](#).

(4) Eurostat. General government total expenditure on education, % of GDP. Source [here](#)

(5) Eurostat. Unemployment by sex and age - annual data. Source [here](#).

(6) Eurostat. Individuals using the internet for finding information about goods and services. Source [here](#).

The data used in our analysis (see [Table 2](#)) reveal clear asymmetries across the EU-27 in both AI uptake and the structural conditions shaping digital development. AI adoption for marketing or sales ranges from just over 1% in Romania to nearly 10% in Sweden, while AI use in production processes spans from values close to 1% in Croatia and Bulgaria to almost 6% in Finland. Similarly, AI deployment in logistics and financial management remains below 1% in several Eastern European countries but reaches above 5–8% in Denmark, Belgium, and Luxembourg.

Non-AI indicators display equally wide disparities. Difficulties in recruiting ICT specialists (diffQual and diffSal) vary from below 1% in Poland to over 10% in Malta, while the share of enterprises employing ICT specialists ranges from 13% in Italy to

35% in Malta. Consumption expenditure patterns differ markedly as well, with ICT-related spending (conICT) ranging from 0.5% of GDP to almost 3%, and energy expenditure showing similarly broad variation. Internet use for finding information about goods and services ranges from only 56% in Romania to over 96% in the Netherlands, illustrating strong differences in digital literacy and consumer digital engagement.

Altogether, these disparities point to a highly heterogeneous digital landscape within the EU-27, where both AI-specific and broader contextual indicators vary substantially across countries. Such complexity reinforces the need for multivariate statistical methods, which are better suited than univariate approaches for capturing the underlying structure of these digital and AI-related asymmetries.

All the variables used have been statistically adjusted in percentages of the enterprises in the country and percentage of usage relative to the country. The percentage of enterprises that use AI technologies for marketing or sales (aiMrk), percentage of enterprises that use AI technologies for production processes (aiProd), percentage of enterprises that use AI technologies for logistics (aiLog), percentage of enterprises that use AI technologies for accounting, controlling or finance management (aiFin) are mainly used to measure the adoption of AI in the major industries of an EU-27 country. These variables reflect the areas where algorithmic decision-making and automation are most systematically integrated into enterprise operations in a country [1] [8]. Skills shortages are one of the strongest predictors of whether firms can adopt AI or advanced digital technologies. Hence, the variables - percentage of enterprises that have difficulty to recruit ICT specialists due to Applicants' lack of relevant ICT qualifications from education and/or training (diffQual), percentage of enterprises that have difficulty to recruit ICT specialists due to applicants' salary expectations too high (diffSal) and percentage of enterprises that employed ICT specialists or recruited/tried to recruit personnel for jobs requiring ICT specialist skills (ictEmp) were considered.

Table 2: Dataset used for analysis.

ID	country	aiMrk	aiProd	aiLog	aiFin	diffQual	diffSal	ictEmp	conEnergy	conICT	conEdu	unemp	intUse
1	Austria	9.64	5.15	0.72	5.62	4.52	3.93	20.45	2.1	1.7	0.5	3.6	77.96
2	Belgium	5.97	5.31	2.64	7.8	6.90	5.77	29.67	2.2	1.6	0.2	3.5	81.37
3	Bulgaria	2.11	1.27	0.51	1.13	3.10	3.11	18.78	2.8	2.8	0.8	2.6	61.85
4	Croatia	3.1	1.16	0.38	1.03	2.60	2.31	16.19	3.3	3.4	0.5	3	73.52
5	Cyprus	3.56	0.54	0.27	1.55	4.68	4.72	29.12	1.9	1.8	1.9	3.5	89.22
6	Czechia	5.95	1.81	0.81	2.41	4.72	4.36	20.75	3.1	1.9	0.2	1.8	86.19
7	Denmark	9.48	4.4	2.28	8.37	6.80	4.67	31.65	2	2.3	0.4	4.6	94.13
8	Estonia	6.97	3.47	1	2.95	3.09	2.50	19.97	4.3	1.9	0.3	5.6	81.24
9	Finland	9.25	5.9	1.83	6.62	3.52	3.40	30.63	2.5	1.7	0.1	5.8	88.54
10	France	2.93	1.88	0.59	1.12	4.21	3.99	16.65	2.3	1.9	0.4	4.8	75.98
11	Germany	6.55	4.85	0.83	4.76	6.92	6.53	23.63	2.5	1.6	0.4	2.4	77.76
12	Greece	1.68	0.89	0.3	1	3.77	3.80	24.43	2.9	3	1.4	6.2	79.41
13	Hungary	2.32	1.4	0.36	1.09	2.58	3.15	29.66	1.5	1.9	0.7	3.1	83.19
14	Ireland	5.63	3.69	1.53	3.4	4.21	3.99	30.59	1.1	0.7	0.3	3.1	93.17
15	Italy	2.93	1.88	0.57	1.12	4.21	3.99	13.21	2.1	1.5	0.5	3.8	58.07
16	Latvia	3.39	1.84	0.88	1.93	1.96	1.93	18.2	2.8	2.4	0.7	4.8	70.72
17	Lithuania	2.85	1.9	0.95	2.28	2.74	2.62	16.77	1.9	2.2	0.3	5.1	79.73
18	Luxembourg	4.27	4.11	1.08	4.72	6.15	6.11	24.25	0.7	0.6	0.3	4.3	72.51
19	Malta	6.51	3.93	1.58	1.24	10.07	10.56	35.18	0.8	1.7	0.9	2.3	80.67
20	Netherlands	8.86	4.68	1.55	5.5	7.77	5.95	30.69	1.7	1.7	0.4	2.8	96.28
21	Poland	2.4	1.84	0.6	1.86	0.91	1.49	25.99	4.3	2.4	0.4	1.8	69.26
22	Portugal	2.86	2.43	0.79	2.13	2.26	3.07	21.6	1.7	1.7	0.9	4.4	77.8
23	Romania	1.13	1.04	0.34	0.64	4.21	3.99	15.57	5.8	2.9	0.7	3.1	56.28
24	Slovakia	3.46	1.86	0.64	2.36	2.75	2.40	17.93	4.1	2.3	0.8	3.6	70.07
25	Slovenia	6.84	5.67	2.59	5.33	2.75	4.01	17.84	2.6	2	0.6	2.4	78.9
26	Spain	3.25	3.03	0.96	2.16	2.42	2.80	20.33	1.7	1.8	0.7	7.4	79.99
27	Sweden	10.02	4.71	1.17	4.06	3.92	2.49	22.19	2.2	2	0.1	6.3	89.35

5. Results

5.1. Factor Analysis

Factor analysis examines the correlations among observed variables to uncover latent structures that explain their common variation [17]. Because this technique relies on the existence of meaningful correlation within the dataset, it is essential to verify that a sufficient correlation structure is present prior to extraction. Our analysis followed the established sequence of methodological steps. First, we inspected the correlation matrix to determine whether the relationships among the AI adoption indicators (AI for marketing, production, logistics, and finance) and the contextual variables (labor-market constraints, ICT employment, macro-consumption patterns, unemployment, and internet use) justified the application of factor analysis. Then we decided to go with the principal component analysis as our extraction method. The factor analysis was chosen as the extraction method because it captures the total variance of the variables including shared (Communality), unique, and error variance since the aim is to summarize the underlying structure of a wide set of indicators [5]. The correlation matrix seen in [Table 3](#) gives evidence of the high correlation coefficient of each variable and at least one another.

Table 3: Correlation matrix

	aiMrk	aiProd	aiLog	aiFin	diffQual	diffSal	ictEmp	conEnergy	conICT	conEdu	unemp	intUse
aiMrk	1.00	0.85	0.64	0.79	0.47	0.29	0.43	-0.27	-0.36	-0.47	0.07	0.65
aiProd		1.00	0.80	0.86	0.44	0.37	0.39	-0.34	-0.52	-0.55	0.07	0.46
aiLog			1.00	0.80	0.41	0.35	0.44	-0.31	-0.36	-0.43	-0.01	0.47
aiFin				1.00	0.40	0.23	0.43	-0.25	-0.38	-0.48	0.06	0.52
diffQual					1.00	0.93	0.56	-0.42	-0.37	-0.06	-0.25	0.33
diffSal						1.00	0.54	-0.47	-0.41	0.09	-0.34	0.21
ictEmp							1.00	-0.48	-0.38	0.06	-0.13	0.68
conEnergy								1.00	0.65	-0.01	-0.12	-0.49
conICT									1.00	0.28	0.00	-0.35
conEdu										1.00	-0.01	-0.12
unemp											1.00	0.17
intUse												1.00

The strongest correlation observed is between Difficulty to recruit ICT specialists (diffQual) and Difficulty to recruit ICT specialists (diffSal) with a correlation coefficient of 0.93. The variables Enterprises use AI technologies for production processes (aiProd) and Enterprises use AI technologies for accounting, controlling or finance management (aiFin), with also a strong positive correlation coefficient of 0.86, such correlation points that these two variables are measuring related aspects of the underlying phenomenon or are influenced by similar factors. The third highest correlation is between Enterprises use AI technologies for marketing or sales (aiMrk) and Enterprises use AI technologies for production processes (aiProd), with a coefficient of 0.85.

Enterprises use AI technologies for logistics (aiLog) and Enterprises use AI technologies for accounting, controlling or finance management (aiFin) also have a solid correlation of 0.80, while slightly lower than the previous two, this is still considered a strong positive correlation. This strong correlation between AI variables suggests a prominent common factor. Our determination of the number of factors to retain (four factors) was using a joint decision rule: Kaiser criterion (eigenvalues > 1), and cumulative variance explained (target $\geq 70\text{--}80\%$) [18]. We applied the Varimax rotation. This is because there was no significant difference between the Varimax and Quartimax rotations, and since Varimax tries to maximize variance of the squared loadings, each factor will have fewer variables with larger loading, facilitating us with the interpretability of factors.

Based on the selected variables, four common factors were created as can be seen in *Table 4*. AI Usage aggregates five of the original variables combining Enterprises use AI technologies for production processes (aiProd), AI usage for finance management (aiFin), for marketing or sales (aiMrk), and logistics (aiLog) along with consumption expenditure for Education services (conEdu).

The second factor, Digital Participation & IT Workforce, combines internet usage (intUse) with Enterprises employed ICT specialists or recruited/tried to recruit personnel for jobs requiring ICT specialist skills (ictEmp).

The third factor states difficulties faced by enterprises to recruit ICT specialists, caused by high salary expectations (diffSal) and insufficient applicant qualifications (diffQual).

The fourth factor represents the national-level investment intensity in ICT services (conICT) as well as energy (conEnergy). This reflects the broader economic infrastructure of the country that supports digital growth.

Table 4: Results of factor analysis

Rotated Factor Pattern					
ID	Variables	AI Usage	Digital Participation & IT Workforce	ICT Employment Shortage	Digital & Energy Investment
1	aiProd	0.88683	0.17806	0.11789	-0.27054
2	aiFin	0.87171	0.30272	0.05813	-0.08764
3	aiMrk	0.81384	0.38638	0.07136	-0.09557
4	aiLog	0.78962	0.27915	0.16795	-0.12727
5	conEdu	-0.77892	0.39176	0.08826	0.10329
6	intUse	0.38182	0.79059	-0.11943	-0.21749
7	ictEmp	0.19439	0.77208	0.32779	-0.23021
8	diffSal	0.09199	0.29275	0.80564	-0.40529
9	diffQual	0.25496	0.35842	0.73383	-0.31948
10	unemp	0.03456	0.1967	-0.78114	-0.19729
11	conICT	-0.31913	-0.02004	-0.11486	0.85205
12	conEnergy	-0.05719	-0.37183	-0.06166	0.83769

5.2. Clustering

After identifying the latent dimensions through factor analysis, we grouped countries using clusters. Hierarchical clustering was preferred over non-hierarchical methods (e.g. k-means) because it's well-suited for exploratory analysis where the underlying group structure is unknown (it does not require pre-specifying the number of clusters), it's also more adequate for small-to-moderate datasets. This method allows the use of various linkage methods and different measures of distance, which contributes to more robust and interpretable clusters based on the data's structure [19]. Hierarchical clustering also allows analysis of R-square values and provides a dendrogram too, which facilitates a visual understanding on how countries group together at different levels of similarity, offering richer insights into the relationships among EU members. The Nonhierarchical clustering method was then used to cluster the observations based on the number of distinct clusters gotten from the hierarchical clustering analysis.

Given the focus of the study, priority was given to the combination of the first factor (AI usage) with the remaining three other factors (Digital Participation & IT Workforce, ICT Employment Shortage, Digital & Energy Investment), in a total of three scatter plots. For each scatter plot, different linkage methods were tested (Single, Centroid, Complete, and Ward), combined with Euclidean distance. We performed an intuitive clustering firstly on all the pairs before performing a cluster analysis on the variables and the respective observations. Ward's method showed the best results and was selected for the cluster analysis. We will now review each of these three scatter plots selected for our study.

5.2.1. AI Usage vs Digital Participation & ICT workforce

Figure 1 shows the distribution of countries according to their standardized scores on two latent dimensions extracted through factor analysis: Factor 1 (Digital Participation & ICT Workforce) on the horizontal axis and Factor 2 (AI Usage) on the vertical axis. The plot visualizes how countries cluster based on shared underlying structures in the data, allowing us to observe comparative strengths and weaknesses beyond individual indicators.

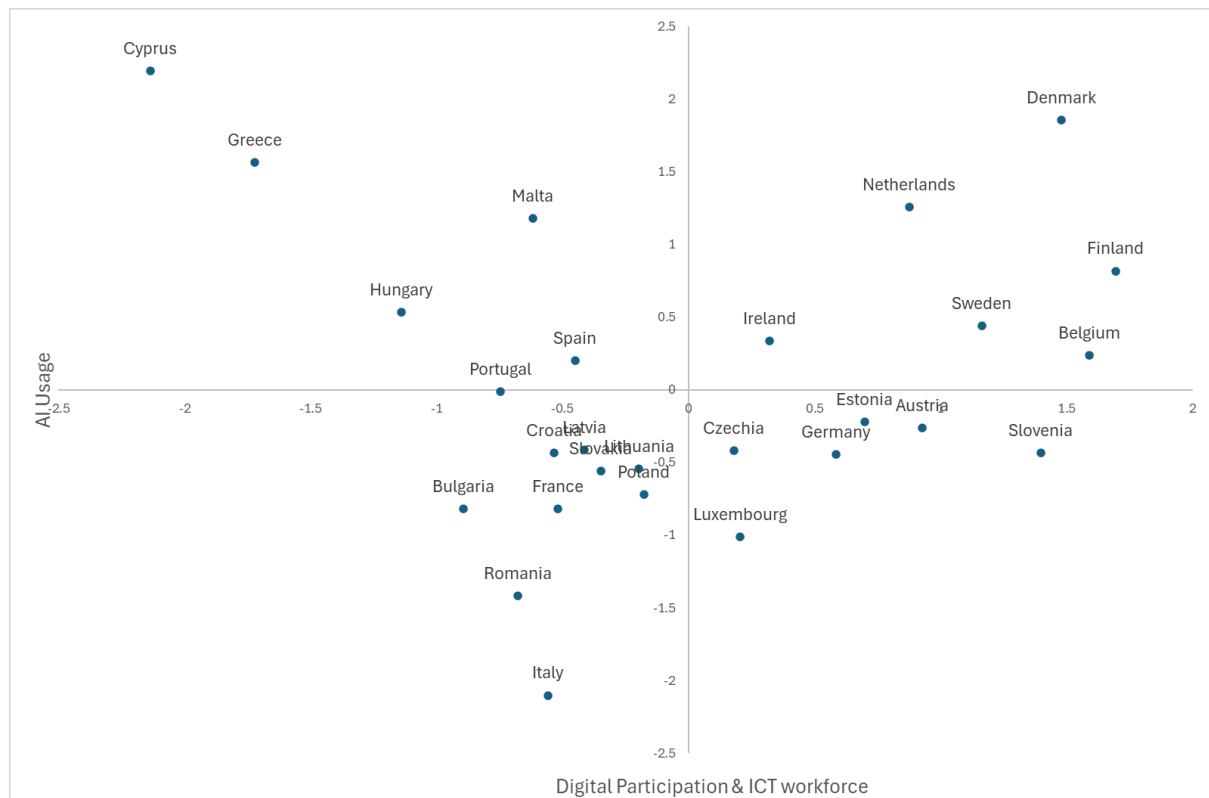


Figure 1: Countries coordinates on factors (AI Usage vs Digital Participation & ICT workforce scatter plot)

The analysis of the cluster formation in *figure 1* gave us some noticeable deductions among the EU-27 member states. In examining the scree plot intuitively along the axis of both factors, six distinct groups emerge. At the lower end of both dimensions, Bulgaria, Romania, Italy, and France form the ***Digitally Laggard Economies***, reflecting limited AI uptake alongside weak ICT workforce participation. Croatia, Latvia, Slovakia, Lithuania and Poland form a ***Digitally Slow Economies***, representing their weak participation in both AI adoption and Digital participation and ICT workforce but not as low as the Digitally Laggard group. Malta, Hungary, Portugal and Spain make up the ***Emerging Digital Economies***, this is because they are positioned at moderate levels on the axis of digital labor capacity but still a bit delayed on their AI adoption. A seemingly large unbalanced profile is found in Cyprus and Greece, where they exhibit higher digital participation yet low AI usage; we would term these countries as the ***Peripheral Digital Economies***, indicating societal capabilities that is not yet matched by enterprise-driven digitalization and adoption of modern technological trend.

Also, Ireland, Czechia, Germany, Estonia, Austria, Slovenia and Luxembourg sit near the centre of both axes, representing the **Balanced Digital Economies** with stable digital maturity across firms and society and developing AI adoption. At the top right end of both axes, we have Denmark, Netherlands, Finland, Sweden and Belgium. These constitute the **Digital Leaders**, combining strong AI adoption with a highly developed ICT workforce and broad societal digital participation. These six clusters collectively seem to intuitively illustrate the multidimensional nature of digital development in the EU 27 states, highlighting how structural labor capacity and enterprise adoption interact with national digital trajectories. However, correlation is not causality and as such, we cannot certainly provide explanations for the disparity between the countries.

The hierarchical method was then carried out and produced the results as can be seen in the dendrogram below (*figure 2*). The horizontal axis measures the distance, and the vertical axis represents the countries. From left to right, the dendrogram maps the clusters' formation.

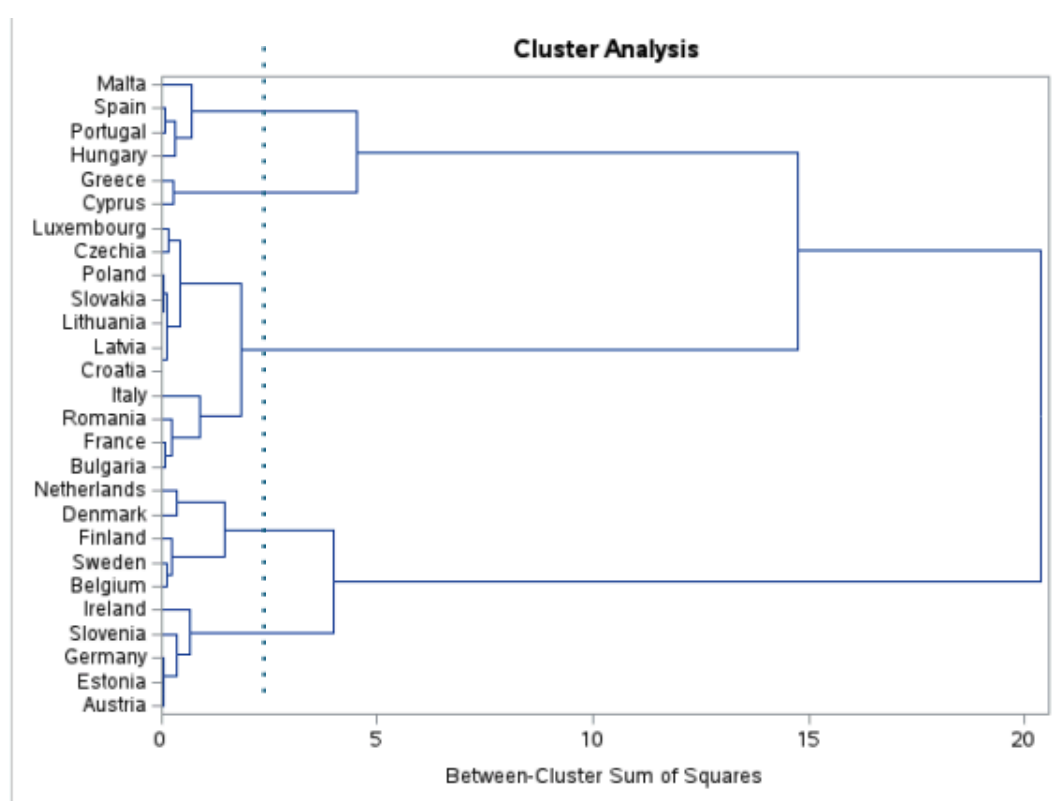


Figure 2: Ward's dendrogram for the AI Adoption on Digital Participation factors across the EU-27.

From the *figure 2* above and the cluster history, we discovered that Croatia and Latvia make up the first pair from the group of countries. This is closely followed by the pair of Lithuania and Slovakia. Following the algorithm of the pairing, all the countries are grouped into their various clusters. Since our final focus was on the non-hierarchical clustering method as stated earlier, the number of clusters to be obtained for the initial seeds was obtained – five clusters. This was as a result of the combination of the observation of the R-square ([Appendix A](#)) and the dendrogram. From the *figure 3*, we can see the different clustering based on the suggestions from the cluster analysis

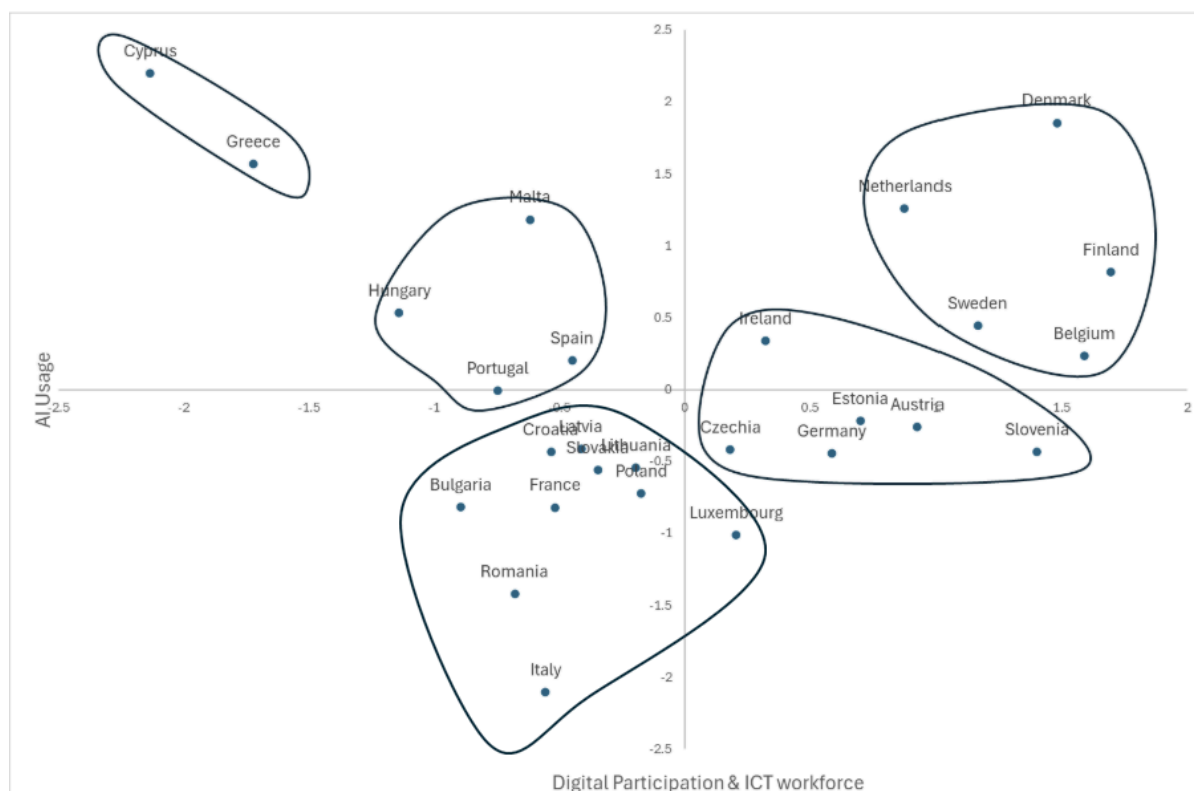


Figure 3: Countries clusters on factors (AI Usage vs Digital Participation & ICT workforce scatter plot)

The cluster solution in *figure 3* above, reveals a different type of clustering. At the top of the trend are the **Digital Leaders**, they remain Denmark, Netherlands, Finland, Sweden, and Belgium, which score high on both AI adoption and digital participation/ICT workforce and thus provide the requisite enabling environment that sustain rapid, digital transformation. The second cluster, the **Balanced Digital Economies** also remained unchanged - (Ireland, Czechia, Germany, Estonia, Austria, Slovenia, Luxembourg). They occupy near-average positions on both dimensions, indicating average digital performance across firms and countries. But contrastingly, we see the **Peripheral Digital Economies**: Cyprus and Greece. They display a distinctive imbalance: comparatively high digital participation among firms coupled with weak AI adoption, suggesting that innovation is not yet embedded in their digital ecosystem and therefore, they are a potentially fragile digital economy.

The remaining two clusters capture intermediate and lagging profiles. The **Emerging Digital Economies**, Portugal, Spain, Malta, Hungary show moderate AI adoption with mixed but generally improving digital participation and ICT workforce capacity, reflecting countries that follow EU norms but still need policy and skills investments to accelerate. Finally, the **Digitally Laggard Economies** (Bulgaria, Romania, Italy, France, Poland, Croatia, Latvia, Slovakia, Lithuania) together combine low AI adoption with weak digital participation and ICT labor market depth, signaling that foundational investments in skills, connectivity and firm capabilities are prerequisites for meaningful advances in AI diffusion.

5.2.2. AI Usage vs ICT Employment Shortage

The visual illustration in *figure 4* reveals eight intuitively emerging clusters, distinguished by their relative positions on AI adoption and ICT employment shortages axes. Germany, Netherlands, Belgium, Slovenia, Denmark, and Austria at the top-right



Figure 4: Countries coordinates on factors (AI Usage vs ICT Employment shortage scatter plot)

form a high-positioned group marked by a great level of ICT shortages combined with mid-to-high level AI adoption, this data. Meanwhile, Estonia, Sweden, and Finland occupy a similar range but extend further right on the horizontal axis, reflecting a higher AI adoption with low labour-market shortages. Czechia, Luxembourg, and Ireland cluster around a center level of both ICT shortages and AI adoption, while Malta stands alone as an extreme outlier with high ICT shortage despite a little above moderate AI adoption.

Looking further, we see, Poland, Slovakia, France, Croatia, and Italy align as mid-level adopters with moderate ICT shortages, forming a central group with balanced digital characteristics. Portugal, Latvia, Lithuania, and Spain sit lower on the ICT-shortage scale while varying modestly in AI adoption, identifying them as lower-pressure, lower-adoption economies. Cyprus and Greece cluster together at the bottom of the vertical axis, representing minimal ICT shortages and low levels of AI adoption. Romania, Bulgaria, and Hungary form another distinct low-to-mid adoption cluster, positioned around moderate ICT shortages but with generally weaker AI uptake, highlighting delayed digitalisation despite some labour-market demand.

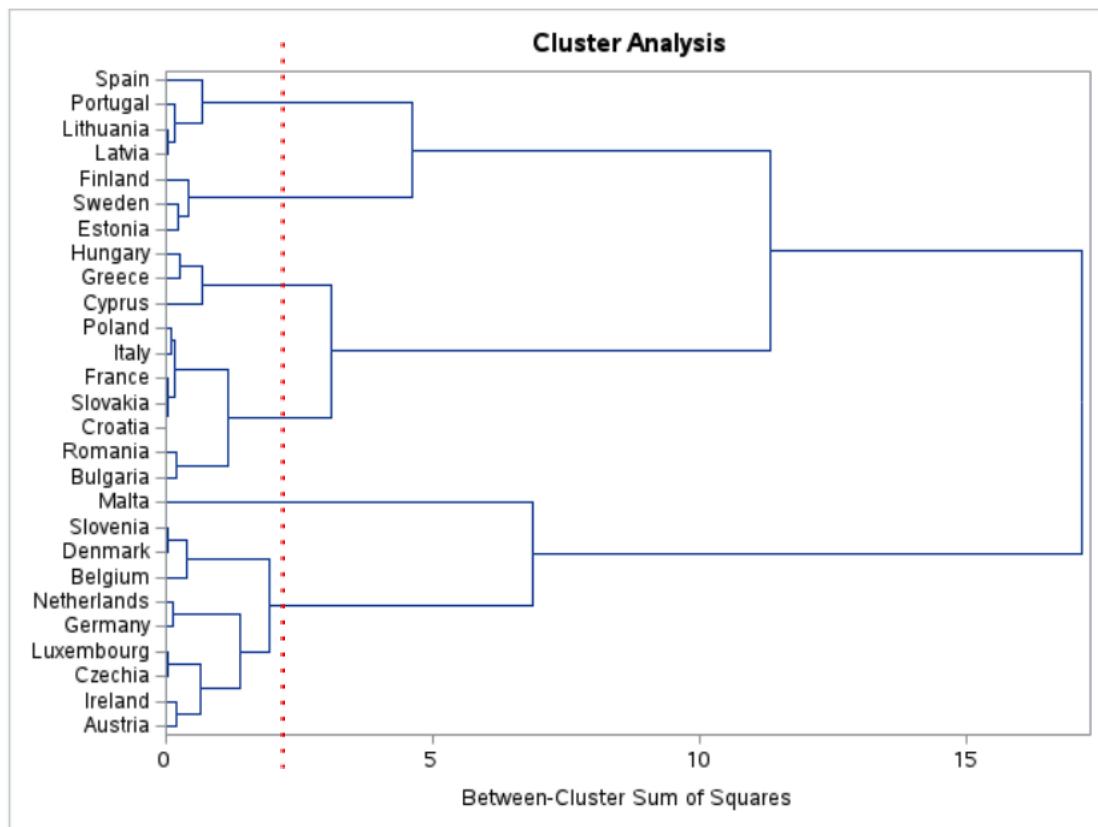


Figure 5: Ward's dendrogram for the AI Adoption on ICT Employment Shortages factors across the EU-27.

From the dendrogram in *figure 5*, and the cluster history, the hierarchical procedure shows a subjective separation when the solution is expanded to six clusters. This provides a differentiation of country profiles when interpreted with ICT employment shortage on the vertical axis and AI adoption on the horizontal axis. The earliest clusters are between Croatia–Slovakia and Denmark–Slovenia, followed by Latvia–Lithuania and other mid-range economies. This eventually forms a larger cluster that includes Poland, Slovakia, France, Croatia, and Italy. Romania, and Bulgaria, merge at a higher distance, indicating a shared lagging profile, and together with earlier merges, complete the six-cluster structure that captures a full spectrum from both factors.

The resulting clusters from the cluster analysis gives more insight to the data in *figure 6*. The six-cluster grouping reflects a representation of digital advancement shaped jointly by AI adoption and ICT employment shortages. At the top of the chart (top-right) of ***digital maturity***, Germany, the Netherlands, Belgium, Slovenia, Denmark, Austria, Czechia, Luxembourg, and Ireland form the biggest cluster. This is characterized by strong AI uptake but also troubled with elevated ICT labor shortages. A second, closely related group, Estonia, Sweden, and Finland, also demonstrate high AI adoption but with comparatively moderate ICT shortages when compared with the earlier cluster, suggesting more ***balanced labor markets*** and more efficient skill development systems. Contrastingly, Romania, Bulgaria, Italy, Croatia, Slovakia, France, and Poland occupy a central position with moderate adoption and moderate shortages. This might represent ***transitioning digital ecosystems*** in which technological diffusion is ongoing but not yet advanced enough to create intense labor-market pressures.

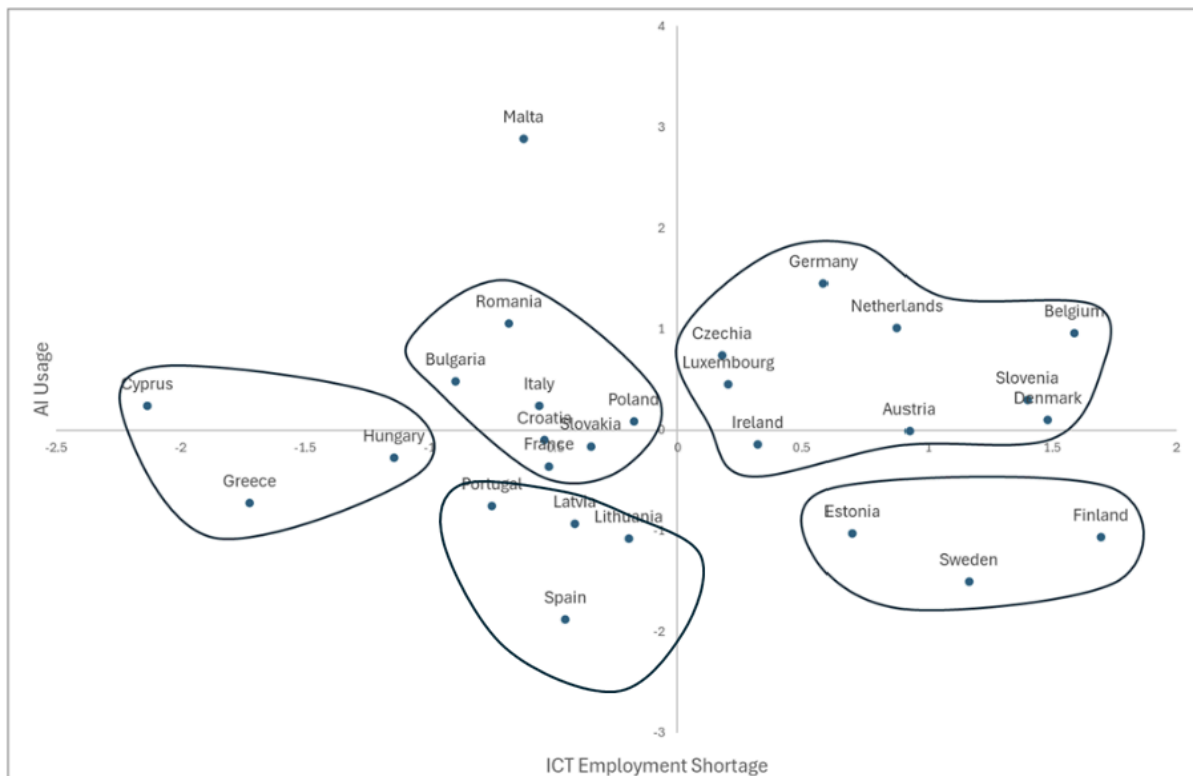


Figure 6: Countries coordinates on factors (AI Usage vs ICT Employment shortage scatter plot)

Toward the negative end of both axes, Portugal, Latvia, Lithuania, and Spain form a cluster of **emerging adopters** whose slower digital evolution is reflected in modest workforce demand. Hungary, Cyprus and Greece lie even further to the left, with the lowest AI uptake and low labour-market pressure, indicative of **limited digital transformation** and weak ICT-intensive activity. Malta stands entirely on its own due to its extreme vertical position, representing the highest ICT workforce shortage in the dataset despite only moderate AI adoption. This unique imbalance isolates Malta from all other groups and underscores a **structural labour-market constraint** not shared by the rest of the EU.

5.2.3. AI Usage vs Digital & Energy Investment

The intuitive clustering derived from the scatterplot reveals a structured gradient of digitalization. Cyprus and Greece form a weak cluster, positioned at the far left and lower end of the graph, indicating both minimal AI adoption and limited investment capacity. A second cluster, Romania, Poland, Croatia, Slovakia, Bulgaria, and Latvia, show significant higher investment in energy and digital contributions despite still modest AI adoption, marking these countries as emerging digital investors whose infrastructural investments are beginning to outpace their technological innovations. In contrast, Lithuania, Hungary, Portugal, France, Malta, Spain, and Italy cluster together in the lower-left region, reflecting low-to-moderate AI adoption combined with restrained investment, situating them as early-stage adopters with slower digital transition momentum.

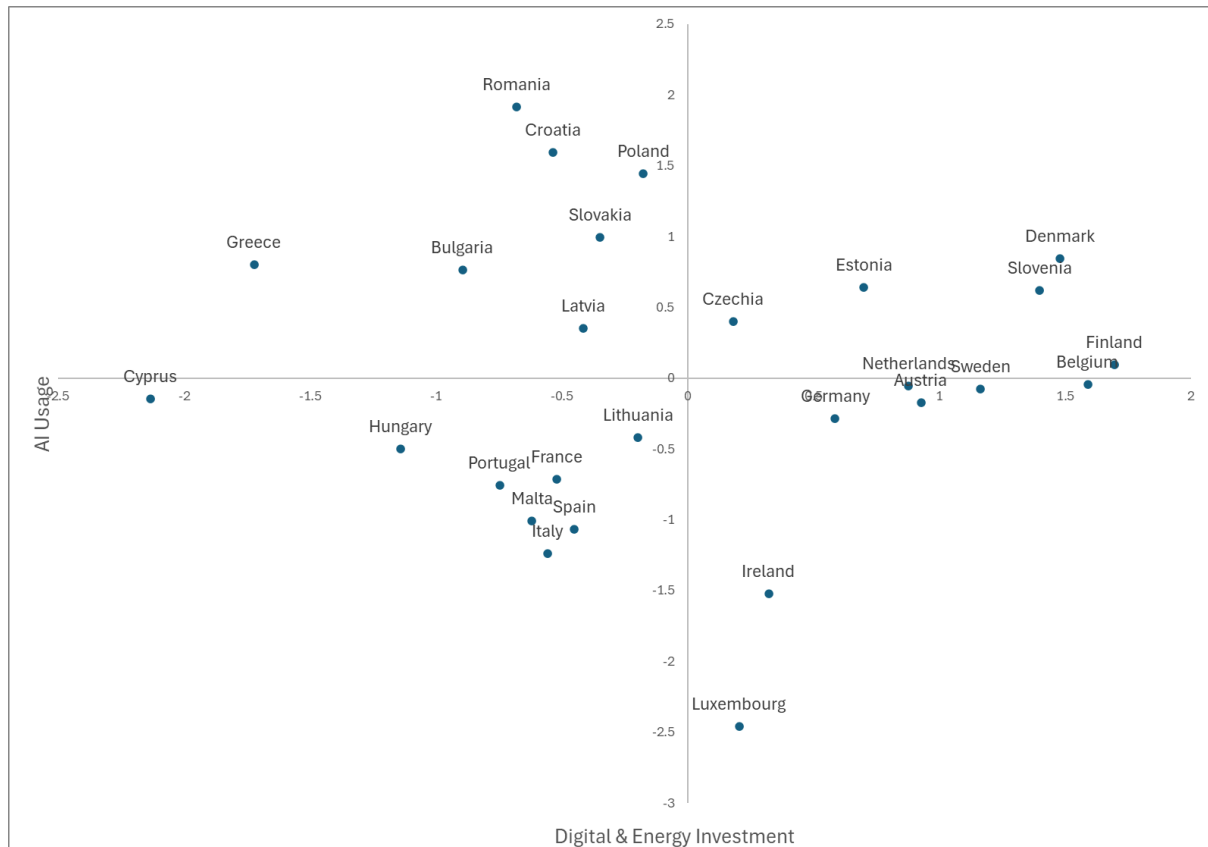


Figure 7: Countries coordinates on factors (AI Usage vs Digital & Energy Investment scatter plot)

A unique pattern also emerges with Ireland and Luxembourg, whose positions in the lower-right quadrant reveal relatively strong AI adoption albeit very low Digital & Energy Investment. This might be suggestive of atypical digital ecosystems driven more by high-value service sectors than by broad national investment. The next cluster is made up of Czechia, Germany, Estonia, the Netherlands, Austria, and Sweden. It aligns toward the right-center of the scatterplot, exhibiting balanced digital development characterized by moderate-to-high AI adoption and correspondingly stable investment profiles. At the top-right frontier, Denmark, Slovenia, Finland, and Belgium occupy the most advanced position, combining high AI adoption with the highest investment levels, identifying them as the leading digital economies where strong financial commitment reinforces rapid technological integration.

After proceeding with the dendrogram as seen in *figure 8*, we first observe that the earliest pairings, Austria with the Netherlands, Belgium with Finland and Malta with Spain reflect strong bilateral similarities that form the foundational building blocks of the later clusters. As the tree continues to develop, additional countries merge into these initial pairs: Italy joins the cluster made up of Malta and Spain. France makes a cluster with Portugal, Denmark with Slovenia and Sweden with the pair of Austria and the Netherlands. By the time the algorithm reaches the point where exactly six clusters remain, these sequential mergers have stabilized into distinct regional and structural configurations. The resulting six clusters are therefore a direct consequence of the hierarchical merging structure rather than predetermined groupings.

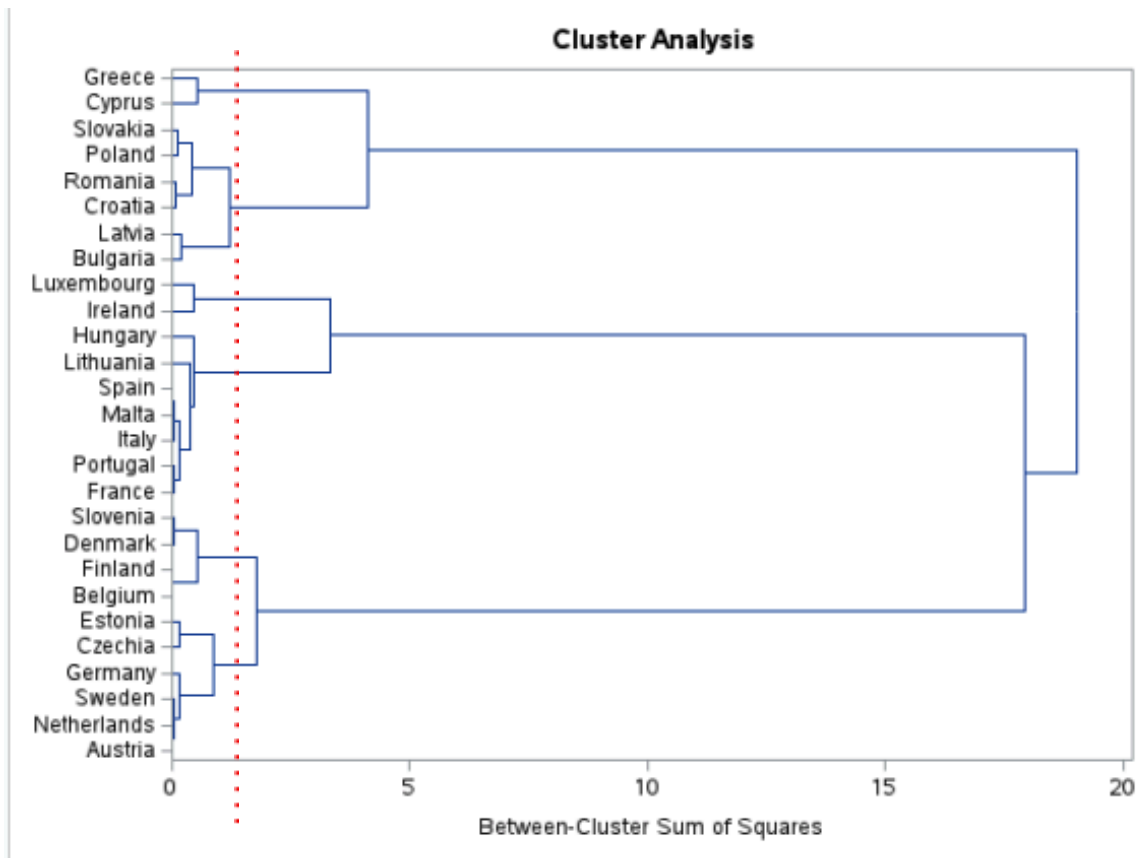


Figure 8: Ward's dendrogram for the AI Adoption on Digital & Energy Investments factors across the EU-27.

The clusters revealed in the scatter plot following the cluster show structural differences in how the EU-27 countries position themselves along the two dimensions of AI adoption and Digital & Energy. The first cluster, formed by Cyprus and Greece, is characterized by comparatively ***low AI adoption and weak investment commitments***. A second cluster that is made up of Romania, Croatia, Poland, Slovakia, Bulgaria, and Latvia, occupies the upper-left portion of the plot, illustrating slightly low AI uptake but higher levels of Digital & Energy Investment. This pattern may suggest economies amid an ***infrastructural acceleration*** whose investment efforts have yet to translate into advanced innovative implementation of which AI is included. The third cluster, consisting of Hungary, Portugal, France, Malta, Spain, Italy, and Lithuania, is situated in the lower-left quadrant and represents countries with similarly modest AI engagement but more moderate investment activity, indicating a ***transitional digital*** profile in which resource allocation remains insufficient to generate momentum comparable to that of the second cluster.

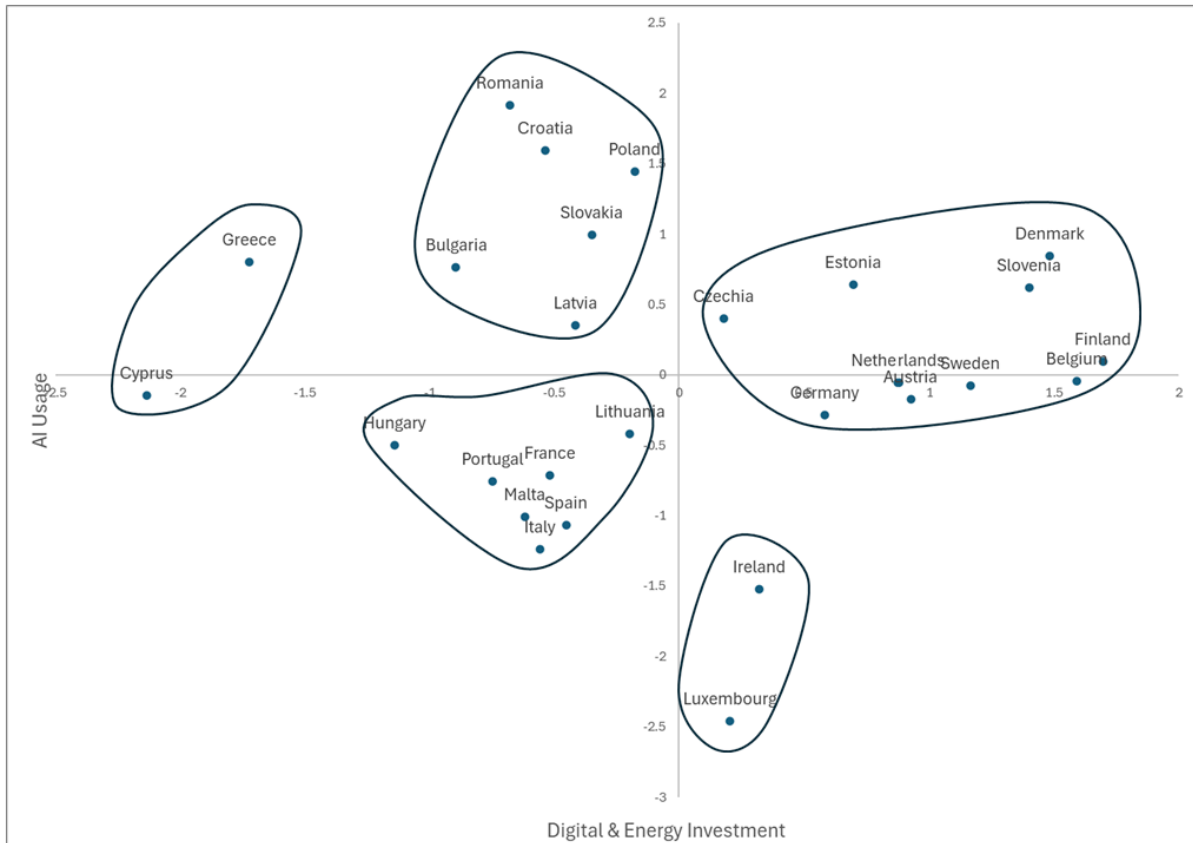


Figure 9: Countries coordinates on factors (AI Usage vs Digital & Energy Investment scatter plot)

The remaining three clusters reflect progressively stronger digital and investment capacities. Ireland and Luxembourg form their own cluster, positioned distinctly in the lower-right quadrant, a mix that combines relatively strong AI adoption with notably low Digital & Energy Investment—likely driven by sector-specific or foreign-led innovation rather than broad national investment agendas. A larger and more cohesive cluster can be found in Czechia, Estonia, Germany, the Netherlands, Austria, Sweden, Finland, Belgium, Slovenia, and Denmark which occupies the right-central to upper-right areas of the graph, marking countries where investment levels and AI uptake are jointly high, supporting an integrated technological and infrastructural ecosystem.

6. Discussion

The purpose of our study was not to explain the causes of artificial intelligence adoption across European Union enterprises, but rather to explore how AI usage appears alongside other digital, labour-market, and investment-related characteristics at the country level. Using principal component factor analysis and cluster analysis allowed us to look at these relationships jointly and to observe how

countries position themselves when several indicators are considered at the same time.

The findings of the conducted factor analysis suggested that AI adoption does not emerge as an isolated phenomenon. Instead, it appears embedded within broader digital and structural dimensions. The extracted factors combine indicators related to AI usage, digital participation and ICT employment, labour-market constraints, and digital and energy investment. We believe the idea that digital transformation processes tend to unfold across multiple, interconnected domains rather than along a single axis.

When these countries were projected into the factor space, the differences in positioning became visible depending on the dimension considered. In particular, countries with higher levels of AI usage tend to be located in the same regions of the factor space as those with stronger digital participation and a larger ICT workforce. At the same time, other countries appear positioned further away, often associated with labour-market shortages or lower investment indicators. These observed patterns suggest that AI adoption is situated within a broader digital and economic context and is shaped by combinations of characteristics rather than by a single dominant factor.

The cluster analysis that was done actually supports this interpretation by grouping countries according to their overall profiles across the extracted factors. The resulting clusters reflect similarities in digital readiness and labour-market conditions, rather than clear stages of development or linear trajectories. We have seen some clusters group countries that display relatively balanced digital environments, while others include countries where constraints in specific dimensions remain more visible. Importantly, the clustering highlights relative similarity within these groups rather than sharp separation between them, which aligns with the continuous nature of the factor space we have observed in the scatter plots.

Looking across the different projections, it is also evident that a country's position varies depending on the factor combination considered, where a country that appears well positioned in terms of digital participation may not occupy the same relative position when labour-market constraints or investment dimensions are considered. This reinforces the idea that AI adoption should be viewed as a multidimensional phenomenon, where strengths in one area do not automatically compensate for weaknesses in another. The scatter plots had illustrated these trade-offs clearly and helped us explain why countries with similar AI usage levels may still differ substantially in their broader digital profiles.

Coming from a comparative perspective, these findings suggest that discussions around AI adoption at the country level should avoid focusing on single indicators or isolated rankings. From what we observed, associations indicate that AI usage tends to move together with other structural characteristics, yet the analysis does not support any inference regarding the direction of influence among these dimensions. Therefore, the results are better interpreted as descriptive patterns that

reflect how digital, labour-market, and investment conditions coexist across countries, rather than as evidence of causal relationships.

From our point of view, several limitations should be considered when interpreting these findings. The analysis relies on aggregated country-level indicators, which may conceal substantial heterogeneity within countries and across firms. In addition to this, the cross-sectional nature of the data restricts the ability to assess dynamic changes over time. Finally, the exploratory methodology employed prioritized pattern detection and dimensional reduction, and as such does not aim to test formal hypotheses or establish the causal mechanisms.

Overall, our discussion highlights that AI adoption across EU enterprises is closely intertwined with broader digital and structural contexts. By combining factor analysis and clustering, the study provides a structured way of comparing countries and of understanding how different dimensions align or diverge. Instead of offering explanations, these findings help clarify how AI adoption fits within a wider configuration of digital and labour-market characteristics at the country level.

7. Conclusion, implications of findings, and limitations

Our goal was to look at how artificial intelligence is used by enterprises across EU countries using exploratory multivariate analysis tools. As mentioned, the intention was not to explain why AI adoption happens, but rather to observe how it appears together along with other characteristics related to digital activity, and labour markets and investment, and to see whether the countries show a similar or different overall profiles.

Looking across all the analyses, AI adoption did not appear evenly distributed across these countries, and the differences observed are not isolated. Countries that scored higher on digital participation and ICT employment were often found in the same part of the factor space as higher AI usage, while countries with weaker digital indicators tend to be positioned differently (Fig. 3). These patterns did repeat across the factor analysis and the clustering results, and the scatter plots as well (Fig. 3–9). So what emerges is not a single dominant explanation, but a set of associations that move together across countries, which should be interpreted as structural patterns rather than causal relationships [5].

The factor analysis had summarized the original chosen variables into four dimensions that reflect AI usage, digital participation and ICT workforce, ICT employment shortages, and digital and energy investment. These factors combined should be interpreted as the hidden latent constructs that bring together the correlated indicators, and not as directly observed quantities. When we group the countries using these scores, clusters form that reflect similarities in their total digital and labour-market context. In this context, clustering highlighted common country

profiles rather than distinct stages or paths of development as seen in scatter plots (Fig. 3–9).

When checking the scatter plots, we find that the position of countries differs and it depends on the dimension considered. Some countries combine relatively high AI usage with strong digital participation (Fig. 3), while other countries show lower AI usage alongside labour-market constraints or lower investment levels (Fig. 6 and Fig. 9). These configurations suggest that AI adoption is situated within a broader context and does not appear independently of other structural characteristics. Yet at the same time, the analysis does not indicate which dimension precedes or determines another, and no direction of influence can be inferred from these results [16].

Seeing this from a policy point of view, the findings suggest that AI adoption should not be discussed as isolation from wider digital and labour-market conditions. And the fact that higher AI usage appears together with certain digital and employment characteristics does not imply that changes in one area will automatically translate into changes in another. But instead, our results underline the importance of considering these dimensions jointly when comparing countries, especially when working with aggregated national data where the patterns are observed in the scatter plots (Fig. 3–9) [15].

We believe that some limitations should be noted. The analysis is based on a cross-section of countries and uses aggregate indicators, which do not reflect firm-level differences within these countries. In addition, the exploratory nature of the methodology we applied means that the results are descriptive and do not support causal interpretation [5, 16]. These limits frame the conclusions that can be drawn from the analysis.

Overall, our project provides a structured descriptive overview of AI adoption across EU enterprises and shows that AI usage is observed alongside broader digital, labour-market and investment characteristics at the country level (Fig. 3–9). And instead of offering explanations, the analysis helps clarify how countries differ and how these differences are organised across several related dimensions.

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9. Computational attachments

Provided in project submission file.